Materials Science – Team Project

<u>Subject:</u>

Heat Treatment and Modeling for Precipitation-Hardenable Aluminum Alloys

1. Background and Motivation

Precipitation-hardenable aluminium alloys (such as EN AW-6061, 6014, or 7075) obtain their mechanical strength from finely dispersed, metastable precipitates formed during heat treatment. The sequence of solution heat treatment, quenching, and artificial ageing determines the type, size, and distribution of these precipitates and thereby the mechanical properties of the alloy.

The microstructural evolution and resulting hardness depend sensitively on three main parameters:

1. Solution heat treatment temperature and duration – affecting the dissolution of alloying elements and the amount of retained secondary phases.
2. Ageing temperature.
3. Ageing duration – controlling precipitation kinetics, coherency, and overageing.

The goal of this project is to experimentally determine and model the influence of these three variables on hardness and microstructure, thereby identifying the optimal heat treatment conditions.

2. Objectives

- Determine the influence of solution heat treatment temperature, solution treatment duration, ageing temperature, and ageing duration on hardness and microstructure.
- Construct a 3D hardness response surface as a function of the three parameters ($T_{solution}$, T_{ageing}, t_{ageing}).
- Develop an empirical or semi-empirical model that describes the hardness response surface and the associated precipitation kinetics.
- Correlate hardness trends with microstructural observations (optical microscopy) and/or X-ray diffraction (XRD) results.
- Identify the condition corresponding to peak strength (T6) and discuss mechanisms of underageing and overageing.

3. Literature Review

Review the precipitation sequence and relevant phases for the chosen Al alloy (e.g. β″ → β′ → β for Al-Mg-Si). Collect data from literature on solution heat treatment temperatures, dissolution kinetics, and diffusion-controlled precipitation. Examine models describing age-hardening kinetics (e.g. Johnson-Mehl-Avrami, Kampmann-Wagner, or empirical hardness models).

4. Experimental Design

a) Solution Heat Treatment

Select three solution heat treatment temperatures (e.g. 520 °C, 540 °C, 560 °C). For each temperature, apply two holding times (e.g. 0.5 h and 1.5 h). Record potential microstructural differences (e.g. grain size, residual undissolved particles).

b) Quenching

Quench samples rapidly (water quench) to retain a supersaturated solid solution. Keep all quenching parameters constant to isolate the influence of solution treatment and ageing.

c) Artificial Ageing

For each solution-treated condition, perform artificial ageing at three temperatures (e.g. 140 °C, 160 °C, 180 °C). Measure hardness after multiple times (e.g. 0.5 h, 1 h, 2 h, 4 h, 8 h). This yields a multi-dimensional data set: $H = f(T_{sol}, T_{age}, t_{age})$.

4.1 Hardness Measurements

Measure Vickers hardness (HV10 or HV1) for all samples. Plot hardness as a function of ageing time for each temperature and solution condition. Generate 3D surface plots (hardness vs. T_{ageing} and t_{ageing} for each $T_{solution}$). Combine surfaces into a global response function.

4.2 Model Development

Develop an empirical or semi-empirical model that describes the hardness evolution as a function where parameters A, k, n are fitted for all solution temperatures. Extend the model to include $T_{solution}$ as a continuous variable. Validate the model by comparing predicted and measured hardness values. Optionally, estimate activation energy using an Arrhenius relation.

4.3. Microstructural and Phase Analysis

Prepare samples representing different regions of the 3D surface: underaged, peak-aged, and overaged. Record and analyze optical micrographs (etched state). Discuss visible microstructural changes.

Record XRD diffractograms for selected ageing states. Identify phases, estimate lattice parameter changes, and discuss evidence for precipitation or solute depletion.

4.4. Mechanical Testing

Conduct tensile tests or microhardness mappings on selected conditions (solution-treated + peak-aged + overaged). Compare mechanical properties with hardness and microstructure. Discuss strengthening mechanisms.

4.5 Data Analysis and Discussion

Visualize results as 2D ageing curves, 3D hardness surfaces, and contour plots. Validate the predictive model and discuss physical meaning. Relate deviations to microstructural or kinetic phenomena such as incomplete dissolution or diffusion limitations.

5. Expected Outcomes

Students will produce complete experimental data sets for hardness evolution over multiple solution and ageing conditions, generate and interpret 3D hardness response surfaces, develop a kinetic model linking processing parameters to hardness, and correlate mechanical, microstructural, and phase analysis data to identify the optimal heat treatment regime.

6. Supervision and Resources

Instructor: Prof. Dr.-Ing. habil. Thorsten Halle

necessary equipment: High-temperature furnace, water quenching station, hardness tester, optical microscope, X-ray diffractometer, and optional tensile testing machine.